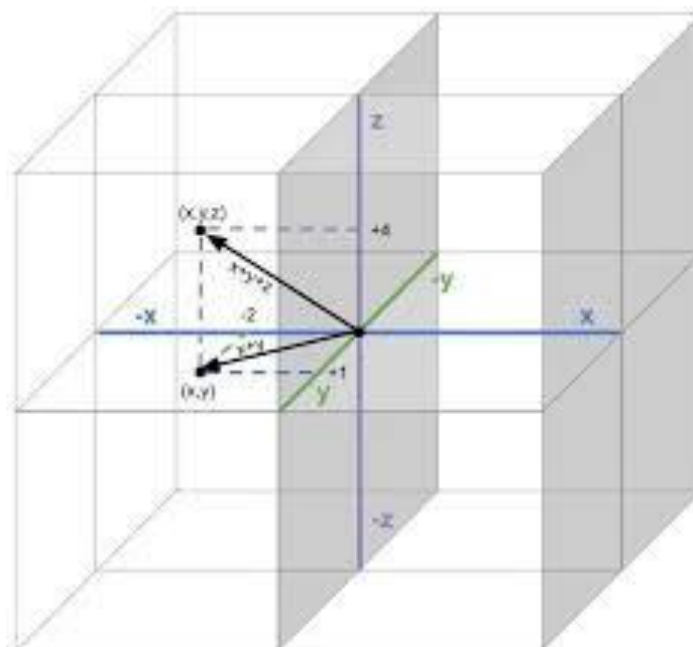


Linear Algebra

Math 1057

Instructor

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15 Linear Maps

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Lecture 15

Linear Maps

Before we begin this Lecture, we review subspaces. Recall that W is a **subspace** of a vector space V if W is a subset of V and

1. the zero vector $\mathbf{0}$ in V is also in W ,
2. for any vectors $\mathbf{u}, \mathbf{v}$ in W the sum $\mathbf{u} + \mathbf{v}$ is also in W , and
3. for any vector $\mathbf{u}$ in W and any scalar α the vector $\alpha\mathbf{u}$ is also in W .

In the previous lecture we gave several examples of subspaces. For example, we showed that a line through the origin in $\mathbb{R}^2$ is a subspace of $\mathbb{R}^2$ and we gave examples of subspaces of $\mathbb{P}_n[t]$ and $M_{n \times m}$. We also showed that if $\mathbf{v}_1, \dots, \mathbf{v}_p$ are vectors in a vector space V then

$$W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

is a subspace of V .

15.1 Linear Maps on Vector Spaces

In Lecture 7, we defined what it meant for a vector mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ to be a **linear mapping**. We now want to introduce linear mappings on general vector spaces; you will notice that the definition is essentially the same but the key point to remember is that the underlying spaces are not $\mathbb{R}^n$ but a general vector space.

Definition 15.1: Let $T : V \rightarrow U$ be a mapping of vector spaces. Then T is called a **linear mapping** if

- for any $\mathbf{u}, \mathbf{v}$ in V it holds that $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$, and
- for any scalar α and $\mathbf{u}$ in V it holds that $T(\alpha\mathbf{u}) = \alpha T(\mathbf{u})$.

Example 15.2. Let $V = M_{n \times n}$ be the vector space of $n \times n$ matrices and let $T : V \rightarrow V$ be the mapping

$$T(\mathbf{A}) = \mathbf{A} + \mathbf{A}^T.$$

Is T is a linear mapping?

Solution. Let $\mathbf{A}$ and $\mathbf{B}$ be matrices in V . Then using the properties of the transpose and regrouping we obtain:

$$\begin{aligned}\mathsf{T}(\mathbf{A} + \mathbf{B}) &= (\mathbf{A} + \mathbf{B}) + (\mathbf{A} + \mathbf{B})^T \\ &= \mathbf{A} + \mathbf{B} + \mathbf{A}^T + \mathbf{B}^T \\ &= (\mathbf{A} + \mathbf{A}^T) + (\mathbf{B} + \mathbf{B}^T) \\ &= \mathsf{T}(\mathbf{A}) + \mathsf{T}(\mathbf{B}).\end{aligned}$$

Similarly, if α is any scalar then

$$\begin{aligned}\mathsf{T}(\alpha\mathbf{A}) &= (\alpha\mathbf{A}) + (\alpha\mathbf{A})^T \\ &= \alpha\mathbf{A} + \alpha\mathbf{A}^T \\ &= \alpha(\mathbf{A} + \mathbf{A}^T) \\ &= \alpha\mathsf{T}(\mathbf{A}).\end{aligned}$$

This proves that T satisfies both conditions of Definition 15.1 and thus T is a linear mapping. $\square$

Example 15.3. Let $\mathsf{V} = M_{n \times n}$ be the vector space of $n \times n$ matrices, where $n \geq 2$, and let $\mathsf{T} : \mathsf{V} \rightarrow \mathbb{R}$ be the mapping

$$\mathsf{T}(\mathbf{A}) = \det(\mathbf{A})$$

Is T is a linear mapping?

Solution. If T is a linear mapping then according to Definition 15.1, we must have $\mathsf{T}(\mathbf{A} + \mathbf{B}) = \det(\mathbf{A} + \mathbf{B}) = \det(\mathbf{A}) + \det(\mathbf{B})$ and also $\mathsf{T}(\alpha\mathbf{A}) = \alpha\mathsf{T}(\mathbf{A})$ for any scalar α . Do these properties actually hold though? For example, we know from the properties of the determinant that $\det(\alpha\mathbf{A}) = \alpha^n \det(\mathbf{A})$ and therefore it does not hold that $\mathsf{T}(\alpha\mathbf{A}) = \alpha\mathsf{T}(\mathbf{A})$ unless $\alpha = 1$. Therefore, T is not a linear mapping. Also, it does not hold in general that $\det(\mathbf{A} + \mathbf{B}) = \det(\mathbf{A}) + \det(\mathbf{B})$; in fact it rarely holds. For example, if

$$\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -1 & 1 \\ 0 & 3 \end{bmatrix}$$

then $\det(\mathbf{A}) = 2$, $\det(\mathbf{B}) = -3$ and therefore $\det(\mathbf{A}) + \det(\mathbf{B}) = -1$. On the other hand,

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 1 & 1 \\ 0 & 4 \end{bmatrix}$$

and thus $\det(\mathbf{A} + \mathbf{B}) = 4$. Thus, $\det(\mathbf{A} + \mathbf{B}) \neq \det(\mathbf{A}) + \det(\mathbf{B})$. $\square$

Example 15.4. Let $\mathsf{V} = \mathbb{P}_n[t]$ be the vector space of polynomials in the variable t of degree no more than $n \geq 1$. Consider the mapping $\mathsf{T} : \mathsf{V} \rightarrow \mathsf{V}$ define as

$$\mathsf{T}(f(t)) = 2f(t) + f'(t).$$

For example, if $f(t) = 3t^6 - t^2 + 5$ then

$$\begin{aligned}\mathsf{T}(f(t)) &= 2f(t) + f'(t) \\ &= 2(3t^6 - t^2 + 5) + (18t^5 - 2t) \\ &= 6t^6 + 18t^5 - 2t^2 - 2t + 10.\end{aligned}$$

Is T is a linear mapping?

Solution. Let $f(t)$ and $g(t)$ be polynomials of degree no more than $n \geq 1$. Then

$$\begin{aligned}\mathsf{T}(f(t) + g(t)) &= 2(f(t) + g(t)) + (f(t) + g(t))' \\ &= 2f(t) + 2g(t) + f'(t) + g'(t) \\ &= (2f(t) + f'(t)) + (2g(t) + g'(t)) \\ &= \mathsf{T}(f(t)) + \mathsf{T}(g(t)).\end{aligned}$$

Therefore, $\mathsf{T}(f(t) + g(t)) = \mathsf{T}(f(t)) + \mathsf{T}(g(t))$. Now let α be any scalar. Then

$$\begin{aligned}\mathsf{T}(\alpha f(t)) &= 2(\alpha f(t)) + (\alpha f(t))' \\ &= 2\alpha f(t) + \alpha f'(t) \\ &= \alpha(2f(t) + f'(t)) \\ &= \alpha \mathsf{T}(f(t)).\end{aligned}$$

Therefore, $\mathsf{T}(\alpha f(t)) = \alpha \mathsf{T}(f(t))$. Therefore, T is a linear mapping. □

We now introduce two important subsets associated to a linear mapping.

Definition 15.5: Let $\mathsf{T} : \mathsf{V} \rightarrow \mathsf{U}$ be a linear mapping.

1. The **kernel** of T is the set of vectors $\mathbf{v}$ in the domain V that get mapped to the zero vector, that is, $\mathsf{T}(\mathbf{v}) = \mathbf{0}$. We denote the kernel of T by $\ker(\mathsf{T})$:

$$\ker(\mathsf{T}) = \{\mathbf{v} \in \mathsf{V} \mid \mathsf{T}(\mathbf{v}) = \mathbf{0}\}.$$

2. The **range** of T is the set of vectors $\mathbf{b}$ in the codomain U for which there exists at least one $\mathbf{v}$ in V such that $\mathsf{T}(\mathbf{v}) = \mathbf{b}$. We denote the range of T by $\text{Range}(\mathsf{T})$:

$$\text{Range}(\mathsf{T}) = \{\mathbf{b} \in \mathsf{U} \mid \text{there exists some } \mathbf{v} \in \mathsf{V} \text{ such that } \mathsf{T}(\mathbf{v}) = \mathbf{b}\}.$$

You may have noticed that the definition of the range of a linear mapping on an abstract vector space is the usual definition of the range of a function. Not surprisingly, the kernel and range are subspaces of the domain and codomain, respectively.

Theorem 15.6: Let $T : V \rightarrow U$ be a linear mapping. Then $\ker(T)$ is a subspace of V and $\text{Range}(T)$ is a subspace of U .

Proof. Suppose that $\mathbf{v}$ and $\mathbf{u}$ are in $\ker(T)$. Then $T(\mathbf{v}) = \mathbf{0}$ and $T(\mathbf{u}) = \mathbf{0}$. Then by linearity of T it holds that

$$T(\mathbf{v} + \mathbf{u}) = T(\mathbf{v}) + T(\mathbf{u}) = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Therefore, since $T(\mathbf{u} + \mathbf{v}) = \mathbf{0}$ then $\mathbf{u} + \mathbf{v}$ is in $\ker(T)$. This shows that $\ker(T)$ is closed under addition. Now suppose that α is any scalar and $\mathbf{v}$ is in $\ker(T)$. Then $T(\mathbf{v}) = \mathbf{0}$ and thus by linearity of T it holds that

$$T(\alpha\mathbf{v}) = \alpha T(\mathbf{v}) = \alpha\mathbf{0} = \mathbf{0}.$$

Therefore, since $T(\alpha\mathbf{v}) = \mathbf{0}$ then $\alpha\mathbf{v}$ is in $\ker(T)$ and this proves that $\ker(T)$ is closed under scalar multiplication. Lastly, by linearity of T it holds that

$$T(\mathbf{0}) = T(\mathbf{v} - \mathbf{v}) = T(\mathbf{v}) - T(\mathbf{v}) = \mathbf{0}$$

that is, $T(\mathbf{0}) = \mathbf{0}$. Therefore, the zero vector $\mathbf{0}$ is in $\ker(T)$. This proves that $\ker(T)$ is a subspace of V . The proof that $\text{Range}(T)$ is a subspace of U is left as an exercise. $\square$

Example 15.7. Let $V = M_{n \times n}$ be the vector space of $n \times n$ matrices and let $T : V \rightarrow V$ be the mapping

$$T(\mathbf{A}) = \mathbf{A} + \mathbf{A}^T.$$

Describe the kernel of T .

Solution. A matrix $\mathbf{A}$ is in the kernel of T if $T(\mathbf{A}) = \mathbf{A} + \mathbf{A}^T = \mathbf{0}$, that is, if $\mathbf{A}^T = -\mathbf{A}$. Hence,

$$\ker(\mathbf{A}) = \{\mathbf{A} \in M_{n \times n} \mid \mathbf{A}^T = -\mathbf{A}\}.$$

What type of matrix $\mathbf{A}$ satisfies $\mathbf{A}^T = -\mathbf{A}$? For example, consider the case that $\mathbf{A}$ is the 2×2 matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

and $\mathbf{A}^T = -\mathbf{A}$. Then

$$\begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} = \begin{bmatrix} -a_{11} & -a_{12} \\ -a_{21} & -a_{22} \end{bmatrix}.$$

Therefore, it must hold that $a_{11} = -a_{11}$, $a_{21} = -a_{12}$ and $a_{22} = -a_{22}$. Then necessarily $a_{11} = 0$ and $a_{22} = 0$ and a_{12} can be arbitrary. For example, the matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 7 \\ -7 & 0 \end{bmatrix}$$

satisfies $\mathbf{A}^T = -\mathbf{A}$. Using a similar computation as above, a 3×3 matrix satisfies $\mathbf{A}^T = -\mathbf{A}$ if $\mathbf{A}$ is of the form

$$\mathbf{A} = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$$

where a, b, c are arbitrary constants. In general, a matrix $\mathbf{A}$ that satisfies $\mathbf{A}^T = -\mathbf{A}$ is called **skew-symmetric**. $\square$

Example 15.8. Let $\mathbf{V}$ be the vector space of differentiable functions on the interval $[a, b]$. That is, f is an element of $\mathbf{V}$ if $f : [a, b] \rightarrow \mathbb{R}$ is differentiable. Describe the kernel of the linear mapping $\mathsf{T} : \mathbf{V} \rightarrow \mathbf{V}$ defined as

$$\mathsf{T}(f(x)) = f(x) + f'(x).$$

Solution. A function f is in the kernel of T if $\mathsf{T}(f(x)) = 0$, that is, if $f(x) + f'(x) = 0$. Equivalently, if $f'(x) = -f(x)$. What functions f do you know of satisfy $f'(x) = -f(x)$? How about $f(x) = e^{-x}$? It is clear that $f'(x) = -e^{-x} = -f(x)$ and thus $f(x) = e^{-x}$ is in $\ker(\mathsf{T})$. How about $g(x) = 2e^{-x}$? We compute that $g'(x) = -2e^{-x} = -g(x)$ and thus g is also in $\ker(\mathsf{T})$. It turns out that the elements of $\ker(\mathsf{T})$ are of the form $f(x) = Ce^{-x}$ for a constant C . $\square$

15.2 Null space and Column space

In the previous section, we introduced the kernel and range of a general linear mapping $\mathsf{T} : \mathbf{V} \rightarrow \mathbf{U}$. In this section, we consider the particular case of matrix mappings $\mathsf{T}_{\mathbf{A}} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ for some $m \times n$ matrix $\mathbf{A}$. In this case, $\mathbf{v}$ is in the kernel of $\mathsf{T}_{\mathbf{A}}$ if and only if $\mathsf{T}_{\mathbf{A}}(\mathbf{v}) = \mathbf{A}\mathbf{v} = \mathbf{0}$. In other words, $\mathbf{v} \in \ker(\mathsf{T}_{\mathbf{A}})$ if and only if $\mathbf{v}$ is a solution to the homogeneous system $\mathbf{A}\mathbf{x} = \mathbf{0}$. Because the case when T is a matrix mapping arises so frequently, we give a name to the set of vectors $\mathbf{v}$ such that $\mathbf{A}\mathbf{v} = \mathbf{0}$.

Definition 15.9: The **null space** of a matrix $\mathbf{A} \in M_{m \times n}$, denoted by $\text{Null}(\mathbf{A})$, is the subset of $\mathbb{R}^n$ consisting of vectors $\mathbf{v}$ such that $\mathbf{A}\mathbf{v} = \mathbf{0}$. In other words, $\mathbf{v} \in \text{Null}(\mathbf{A})$ if and only if $\mathbf{A}\mathbf{v} = \mathbf{0}$. Using set notation:

$$\text{Null}(\mathbf{A}) = \{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{A}\mathbf{v} = \mathbf{0}\}.$$

Hence, the following holds

$$\ker(\mathsf{T}_{\mathbf{A}}) = \text{Null}(\mathbf{A}).$$

Because the kernel of a linear mapping is a subspace we obtain the following.

Theorem 15.10: If $\mathbf{A} \in M_{m \times n}$ then $\text{Null}(\mathbf{A})$ is a subspace of $\mathbb{R}^n$.

Hence, by Theorem 15.10, if $\mathbf{u}$ and $\mathbf{v}$ are two solutions to the linear system $\mathbf{A}\mathbf{x} = \mathbf{0}$ then $\alpha\mathbf{u} + \beta\mathbf{v}$ is also a solution:

$$\mathbf{A}(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha\mathbf{A}\mathbf{u} + \beta\mathbf{A}\mathbf{v} = \alpha \cdot \mathbf{0} + \beta \cdot \mathbf{0} = \mathbf{0}.$$

Example 15.11. Let $V = \mathbb{R}^4$ and consider the following subset of V :

$$W = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 \mid 2x_1 - 3x_2 + x_3 - 7x_4 = 0\}.$$

Is W a subspace of V ?

Solution. The set W is the null space of the matrix 1×4 matrix $\mathbf{A}$ given by

$$\mathbf{A} = \begin{bmatrix} 2 & -3 & 1 & -7 \end{bmatrix}.$$

Hence, $W = \text{Null}(\mathbf{A})$ and consequently W is a subspace. $\square$

From our previous remarks, the null space of a matrix $\mathbf{A} \in M_{m \times n}$ is just the solution set of the homogeneous system $\mathbf{A}\mathbf{x} = \mathbf{0}$. Therefore, one way to explicitly describe the null space of $\mathbf{A}$ is to solve the system $\mathbf{A}\mathbf{x} = \mathbf{0}$ and write the general solution in parametric vector form. From our previous work on solving linear systems, if the $\mathbf{rref}(\mathbf{A})$ has r leading 1's then the number of parameters in the solution set is $d = n - r$. Therefore, after performing back substitution, we will obtain vectors $\mathbf{v}_1, \dots, \mathbf{v}_d$ such that the general solution in parametric vector form can be written as

$$\mathbf{x} = t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \dots + t_d\mathbf{v}_d$$

where $t_1, t_2, \dots, t_d$ are arbitrary numbers. Therefore,

$$\text{Null}(\mathbf{A}) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_d\}.$$

Hence, the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ form a spanning set for $\text{Null}(\mathbf{A})$.

Example 15.12. Find a spanning set for the null space of the matrix

$$\mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}.$$

Solution. The null space of $\mathbf{A}$ is the solution set of the homogeneous system $\mathbf{A}\mathbf{x} = \mathbf{0}$. Performing elementary row operations one obtains

$$\mathbf{A} \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Clearly $r = \text{rank}(\mathbf{A})$ and since $n = 5$ we will have $d = 3$ vectors in a spanning set for $\text{Null}(\mathbf{A})$. Letting $x_5 = t_1$, and $x_4 = t_2$, then from the 2nd row we obtain

$$x_3 = -2t_2 + 2t_1.$$

Letting $x_2 = t_3$, then from the 1st row we obtain

$$x_1 = 2t_3 + t_2 - 3t_1.$$

Writing the general solution in parametric vector form we obtain

$$\mathbf{x} = t_1 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} + t_2 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t_3 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore,

$$\text{Null}(\mathbf{A}) = \text{span} \left\{ \underbrace{\begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{v}_1}, \underbrace{\begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{v}_2}, \underbrace{\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{\mathbf{v}_3} \right\}$$

You can verify that $\mathbf{A}\mathbf{v}_1 = \mathbf{A}\mathbf{v}_2 = \mathbf{A}\mathbf{v}_3 = \mathbf{0}$. □

Now we consider the range of a matrix mapping $\mathbf{T}_{\mathbf{A}} : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Recall that a vector $\mathbf{b}$ in the co-domain $\mathbb{R}^m$ is in the range of $\mathbf{T}_{\mathbf{A}}$ if there exists some vector $\mathbf{x}$ in the domain $\mathbb{R}^n$ such that $\mathbf{T}_{\mathbf{A}}(\mathbf{x}) = \mathbf{b}$. Since, $\mathbf{T}_{\mathbf{A}}(\mathbf{x}) = \mathbf{A}\mathbf{x}$ then $\mathbf{A}\mathbf{x} = \mathbf{b}$. Now, if $\mathbf{A}$ has columns $\mathbf{A} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n]$ and $\mathbf{x} = (x_1, x_2, \dots, x_n)$ then recall that

$$\mathbf{A}\mathbf{x} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n$$

and thus $\mathbf{A}\mathbf{x} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n = \mathbf{b}$. Thus, a vector $\mathbf{b}$ is in the range of $\mathbf{A}$ if it can be written as a linear combination of the columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ of $\mathbf{A}$. This motivates the following definition.

Definition 15.13: Let $\mathbf{A} \in M_{m \times n}$ be a matrix. The span of the columns of $\mathbf{A}$ is called the **column space** of $\mathbf{A}$. The column space of $\mathbf{A}$ is denoted by $\text{Col}(\mathbf{A})$. Explicitly, if $\mathbf{A} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n]$ then

$$\text{Col}(\mathbf{A}) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}.$$

In summary, we can write that

$$\text{Range}(\mathbf{T}_{\mathbf{A}}) = \text{Col}(\mathbf{A}).$$

and since $\text{Range}(\mathbf{T}_{\mathbf{A}})$ is a subspace of $\mathbb{R}^m$ then so is $\text{Col}(\mathbf{A})$.

Theorem 15.14: The column space of a $m \times n$ matrix is a subspace of $\mathbb{R}^m$.

Example 15.15. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}.$$

Is $\mathbf{b}$ in the column space $\text{Col}(\mathbf{A})$?

Solution. The vector $\mathbf{b}$ is in the column space of $\mathbf{A}$ if there exists $\mathbf{x} \in \mathbb{R}^4$ such that $\mathbf{Ax} = \mathbf{b}$. Hence, we must determine if $\mathbf{Ax} = \mathbf{b}$ has a solution. Performing elementary row operations on the augmented matrix $[\mathbf{A} \ \mathbf{b}]$ we obtain

$$[\mathbf{A} \ \mathbf{b}] \sim \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ 0 & 1 & -5 & -4 & -2 \\ 0 & 0 & 0 & 17 & 1 \end{bmatrix}$$

The system is consistent and therefore $\mathbf{Ax} = \mathbf{b}$ will have a solution. Therefore, $\mathbf{b}$ is in $\text{Col}(\mathbf{A})$. $\square$

After this lecture you should know the following:

- what the null space of a matrix is and how to compute it
- what the column space of a matrix is and how to determine if a given vector is in the column space
- what the range and kernel of a linear mapping is